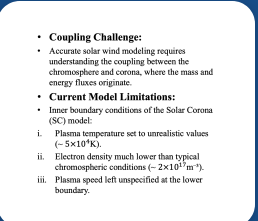


# Toward a Dynamic Chromosphere for AWSoM-R

## A Field-Aligned Two-Fluid Model of Ionization & Chromospheric Evaporation

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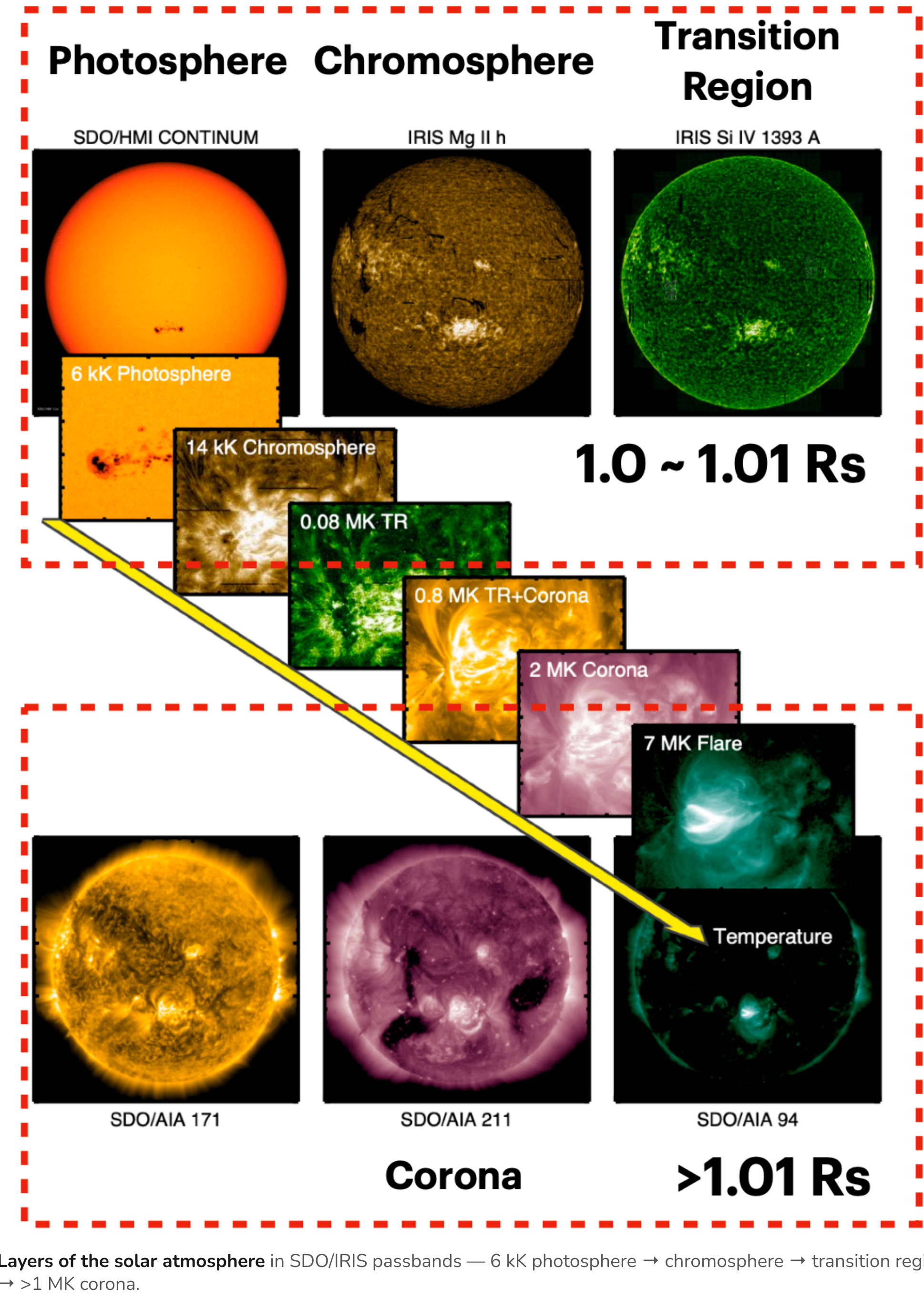
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### Why the chromosphere?

The chromosphere is the **partially-ionized gateway** that sets the mass & energy fed to the corona and solar wind — yet global models **prescribe** it rather than solve it.

- 6 kK → >1 MK across a few thousand km; density drops by orders of magnitude.
- Processes nearly all of the plasma & Poynting flux that fill the corona — via evaporation, spicules, wave heating.

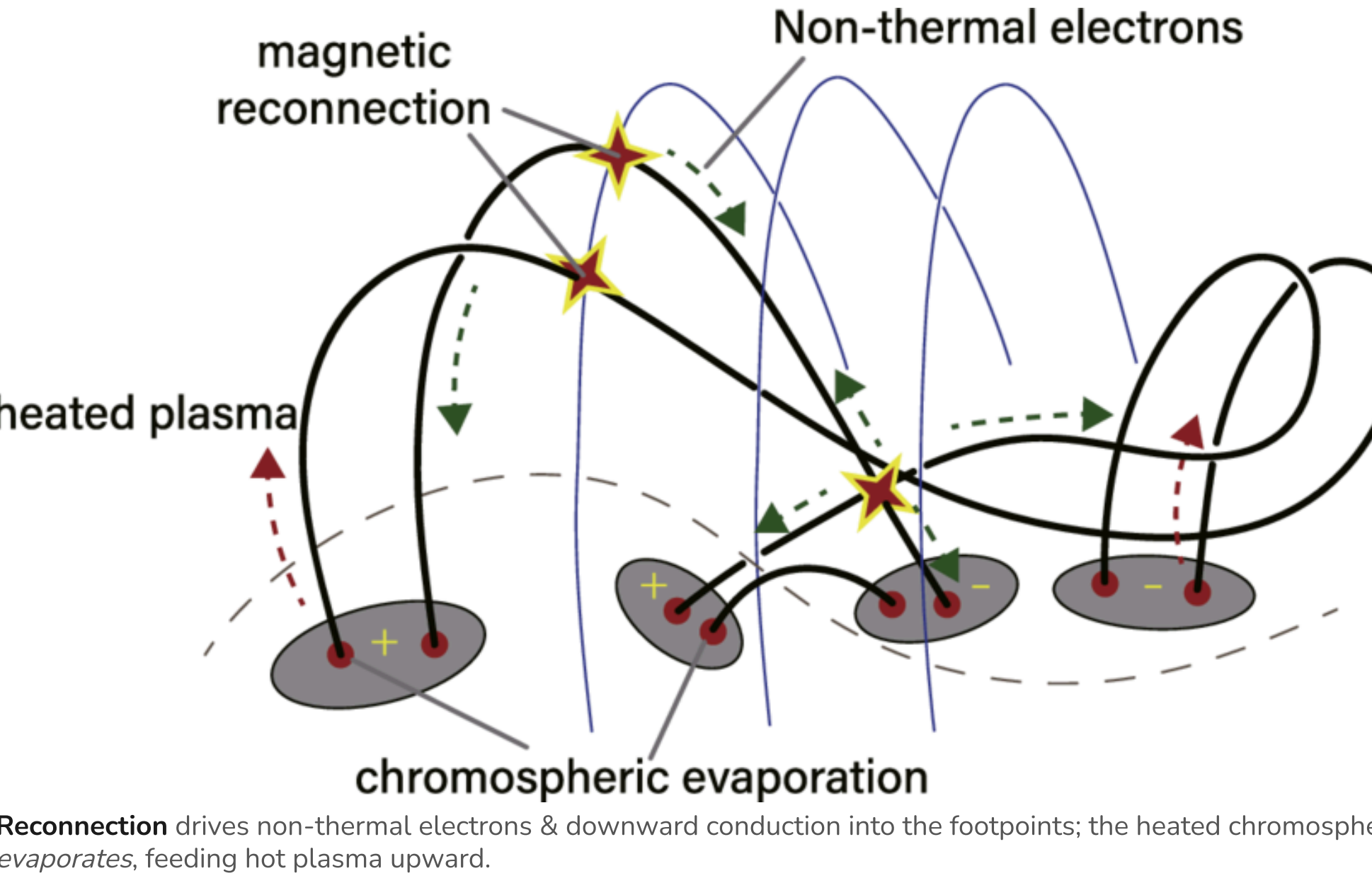


### AWSoM-R's missing lower boundary

**AWSoM / AWSoM-R** places its inner boundary in the upper chromosphere and injects Alfvén-wave Poynting flux through a thin, quasi-static layer. The dynamic chromosphere below is replaced by crude BCs:

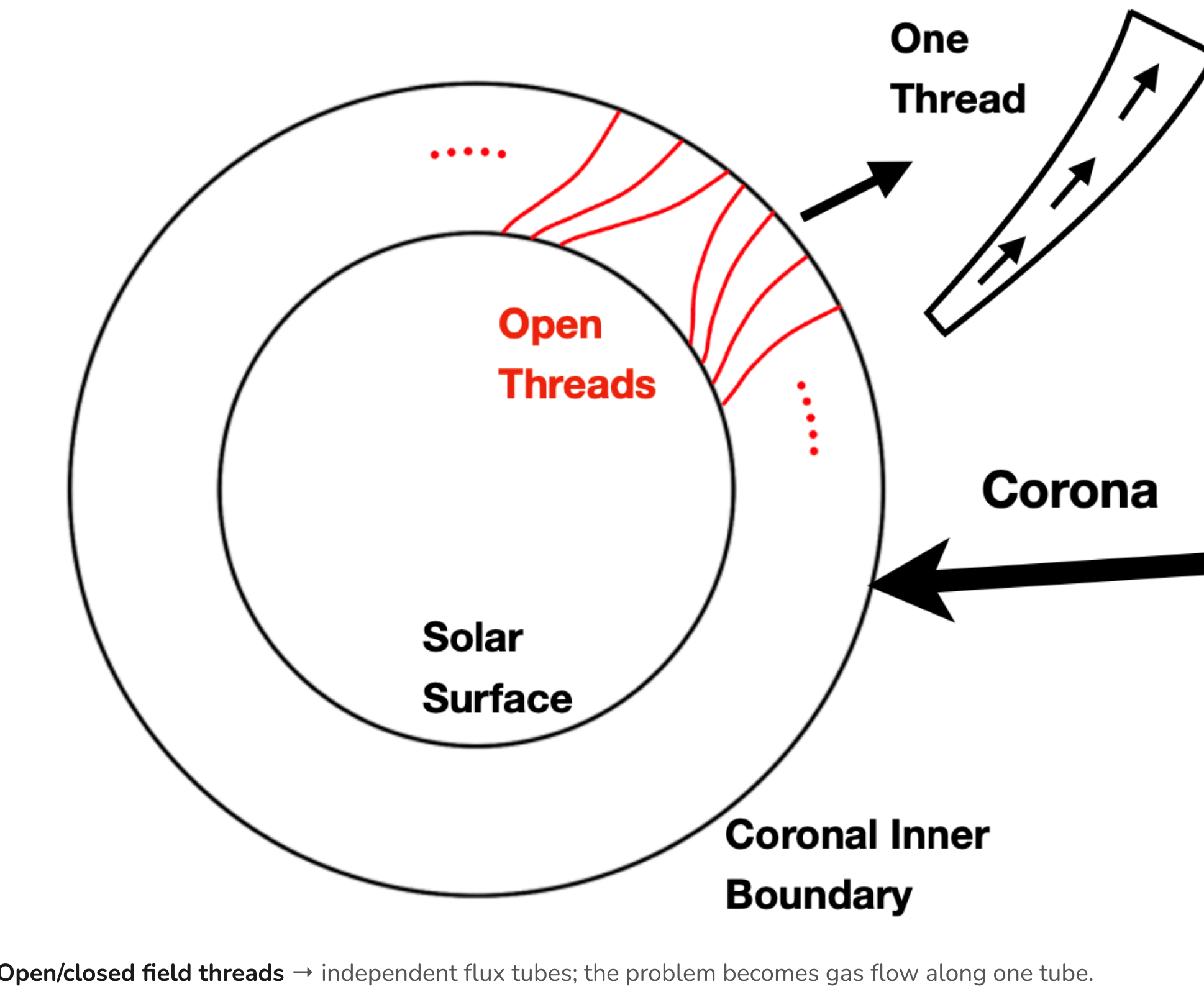
- temperature fixed at unrealistic  $T \sim 5 \times 10^4$  K;
- electron density far below chromospheric,  $n_e \sim 2 \times 10^{17} \text{ m}^{-3}$ ;
- plasma speed **left unspecified**.

**Chromospheric evaporation** — conduction-driven upflows — governs the real mass/energy exchange; it must be **solved, coupled to the corona**.



### From 3D MHD to a 1.5D field-aligned model

- Take  $\mathbf{V} \parallel \mathbf{B}$  with a (PFSS) potential field as a **geometric guide**.
- Split into flux tubes of cross-section  $A(s) \propto 1/B(s)$ ; derivatives reduce to  $\partial/\partial s$ .
- The **B field enters purely as an area factor**.
- 13-variable 3D MHD → **6 field-aligned variables** (7 with a separate  $T_e$ ).



### Two-fluid field-aligned equations

Conservative form (ions  $i$  + neutrals  $n$ ; electrons folded in by quasi-neutrality):

$$\frac{\partial \mathbf{U}}{\partial t} + B \frac{\partial}{\partial s} \left( \frac{\mathbf{F}}{B} \right) = \mathbf{S}$$
$$\mathbf{U} = \begin{bmatrix} \rho_i \\ \rho_n \\ \rho_i V \\ \rho_n U \\ e_i \\ e_n \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho_i V \\ \rho_n U \\ \rho_i V^2 + 2n_i k_B T_i \\ \rho_n U^2 + n_n k_B T_n \\ (e_i + 2n_i k_B T_i) V \\ (e_n + n_n k_B T_n) U \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} +\Gamma \\ -\Gamma \\ -\rho_i g + a_i + R \\ -\rho_n g + a_n - R \\ -\rho_i g V + Q_i \\ -\rho_n g U + Q_n \end{bmatrix}$$

Electron pressure  $p_e = n_i k_B T_e$  enters  $e_i$  via the factor 2. The **source S** couples the fluids and the corona:

- $\Gamma$  — net **ionization** mass exchange (neutral→ion; Voronov, Hummer rates);
- $a_s = p_s B \partial_s(1/B)$  — flux-tube **area force**;  $g$  field-aligned gravity;
- $R = \rho_i \nu_{in}(U - V) + \Gamma U$  — ion–neutral **drag** + ionization momentum;
- $Q_i, Q_n$  — frictional +  $T_i - T_n$  collisional **exchange**, conduction  $-\partial_s q$ , ionization cost, radiative **cooling**.

### Source physics (calibrated to Model C7)

**Non-equilibrium H ionization / recombination** — five channels integrated dynamically:

- collisional  $S_i$  (Voronov 1997);
- multilevel collisional  $S_{CR}$  (detailed balance);
- radiative recomb. (Hummer 1994)  $\alpha_r \propto T_e^{-0.75}$ ;
- three-body  $\alpha_c \propto n_e (k_B T_e)^{-9/2}$  (Hinnov 1962);
- photoionization: **Route-B** C7 inversion ( $h \lesssim 1500$  km) blended to Chae (2021) FAL-C.

**Radiative cooling**: opt-thick H I/Ca II/Mg II (Carlsson & Leenaarts 2012) + opt-thin  $n_e n_H \Lambda(T)$ . **Conduction**: Spitzer  $\kappa_e \propto T_e^{5/2} + \kappa_n$ ; **TRAC** (Johnston 2020). **Optional 3-T**: separate  $T_e \neq T_i$ .

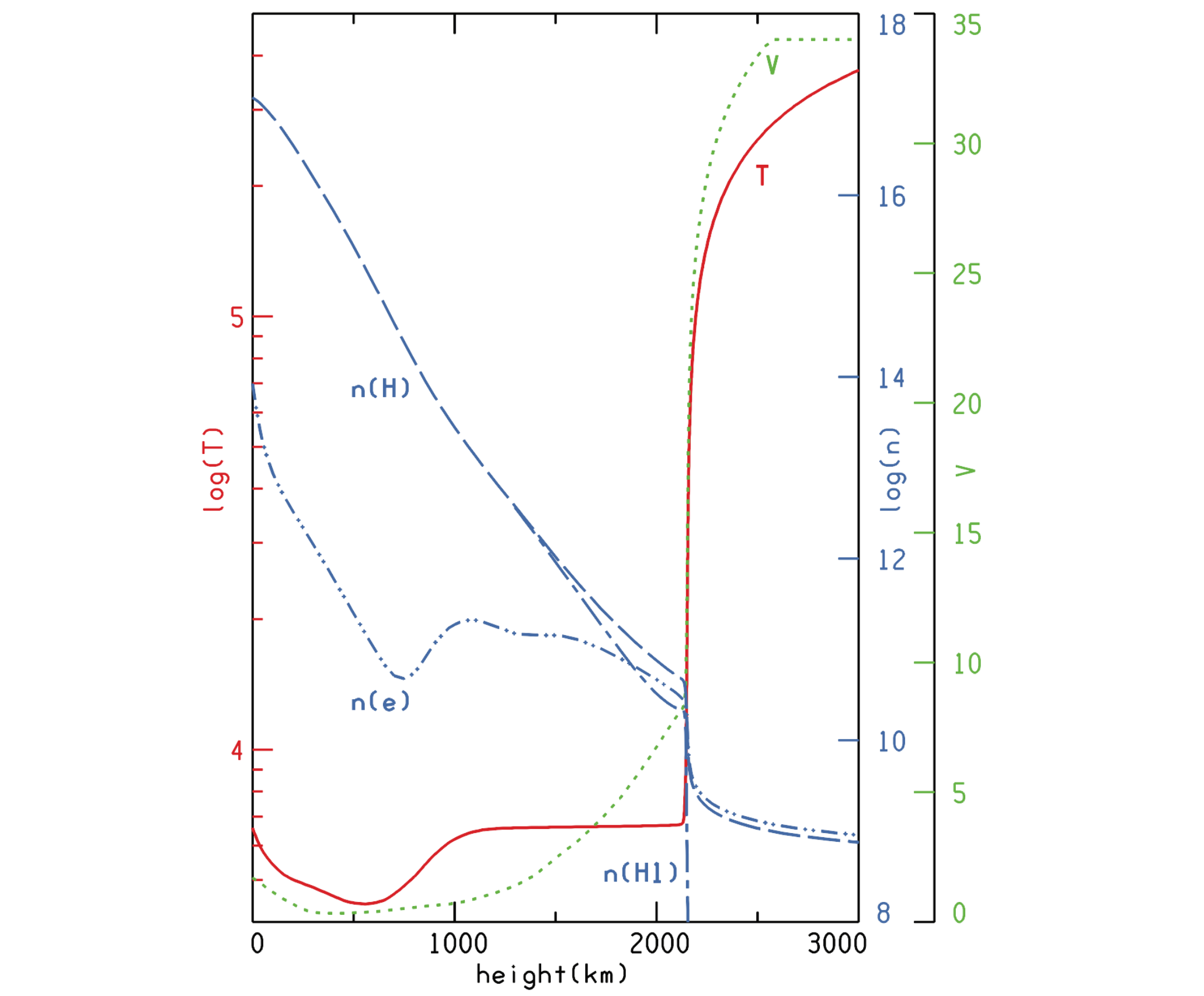


FIG. 8.—Model C7 variations with height of the temperature (in K), turbulent velocity (in  $\text{km s}^{-1}$ ), total hydrogen density, neutral hydrogen density, and electron density (in  $\text{cm}^{-3}$ ).

Model C7 atmosphere (Avrett & Loeser 2008) — the initial state.

### Physically grounded boundary conditions

**Upper (TR/coronal) BC** at  $h \approx 2153$  km,  $T \approx 2.2 \times 10^4$  K (just above the  $\kappa_e/\kappa_n$  crossover). The corona is *outside* the box — its whole influence is three  $T$ -dependent closures on the charge fluid:

- $\rho(T)$  — constant-pressure TR:  $p = (\mu m_H / k_B) P_{\text{TR}} / T \propto 1/T$  (RTV 1978).
- $q(T)$  — downward conductive flux imposed as a **Neumann flux**, not a hot wall:

$$q = -\kappa_0 T^{5/2} \frac{dT}{ds} = \frac{2}{7} \kappa_0 T_{\text{cot}}^{7/2} / L$$

A Dirichlet hot wall over-conducts via  $\kappa_e \propto T^{5/2}$  on the coarse grid → a spurious supersonic downflow. This  $q$  is the **heat that holds the top against ionization + radiative cooling** — load-bearing, paired with the radiative sink.

- $V(T)$  — mass-flux continuity, **Mach-capped**  $|V| \leq 0.05 c_s$  ( $\sim 1$  km/s; Bradshaw & Emslie 2020, Hansteen 1993).
- **Neutrals slaved** — H nearly fully ionized  $\Rightarrow$  single-fluid TR:  $U=V$ ,  $T_n=T$ ,  $\rho_n$  from local equilibrium (no  $\rho_n \rightarrow 0$  blow-up).
- **TRAC** (Johnston 2020) — adaptive cutoff  $T_i$ ; rescale  $\kappa, \Lambda, Q$  to conserve TR-integrated radiation/heating on the coarse grid (eq.-27 jump condition).

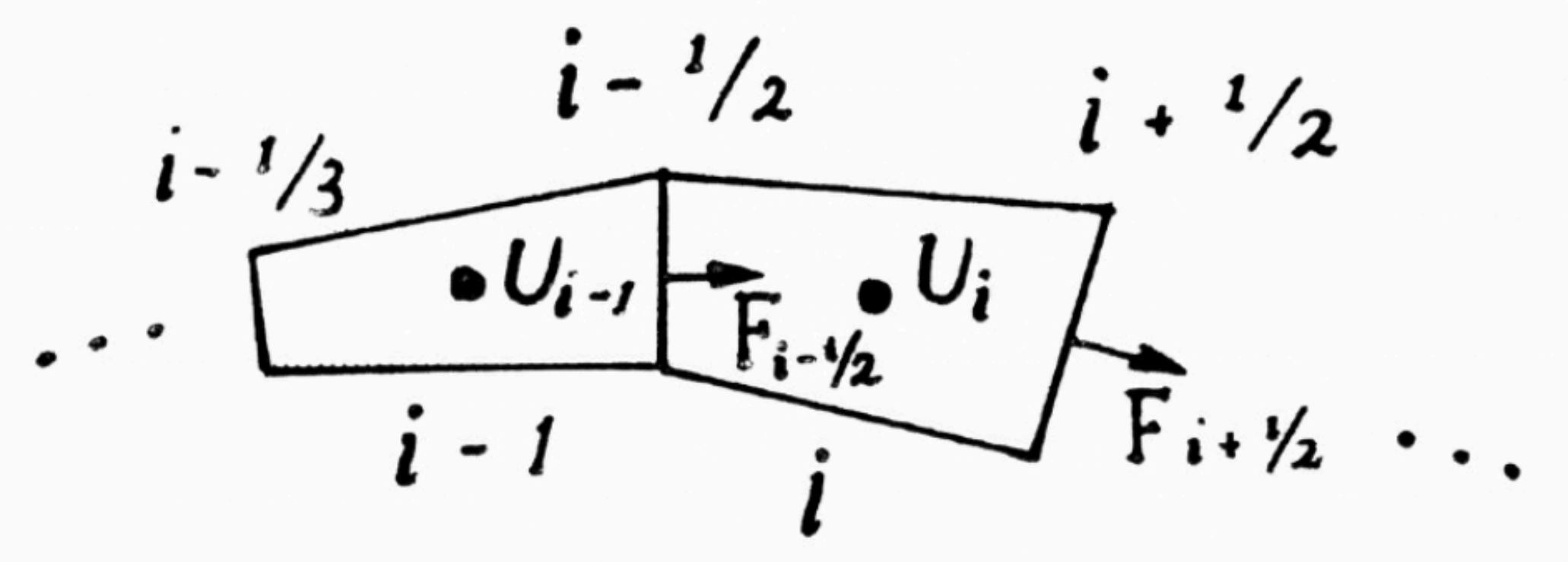
**Lower (photospheric) BC** — a **discrete-HSE** ghost integrates  $dp/ds = -\rho g$  on the *actual* grid, making  $V=U=0$  an **exact fixed point** → removes the  $\sim 25\%$  HSE residual &  $\sim 100$  s base drift (Pandey 2024). Wave-transparent + isotropic numerical diffusion  $\chi \nabla^2$  for TR stability; base  $(n, T)$  fixed to C7.

### Numerical methods

Finite-volume on the field line;  $B$  as a geometric factor:

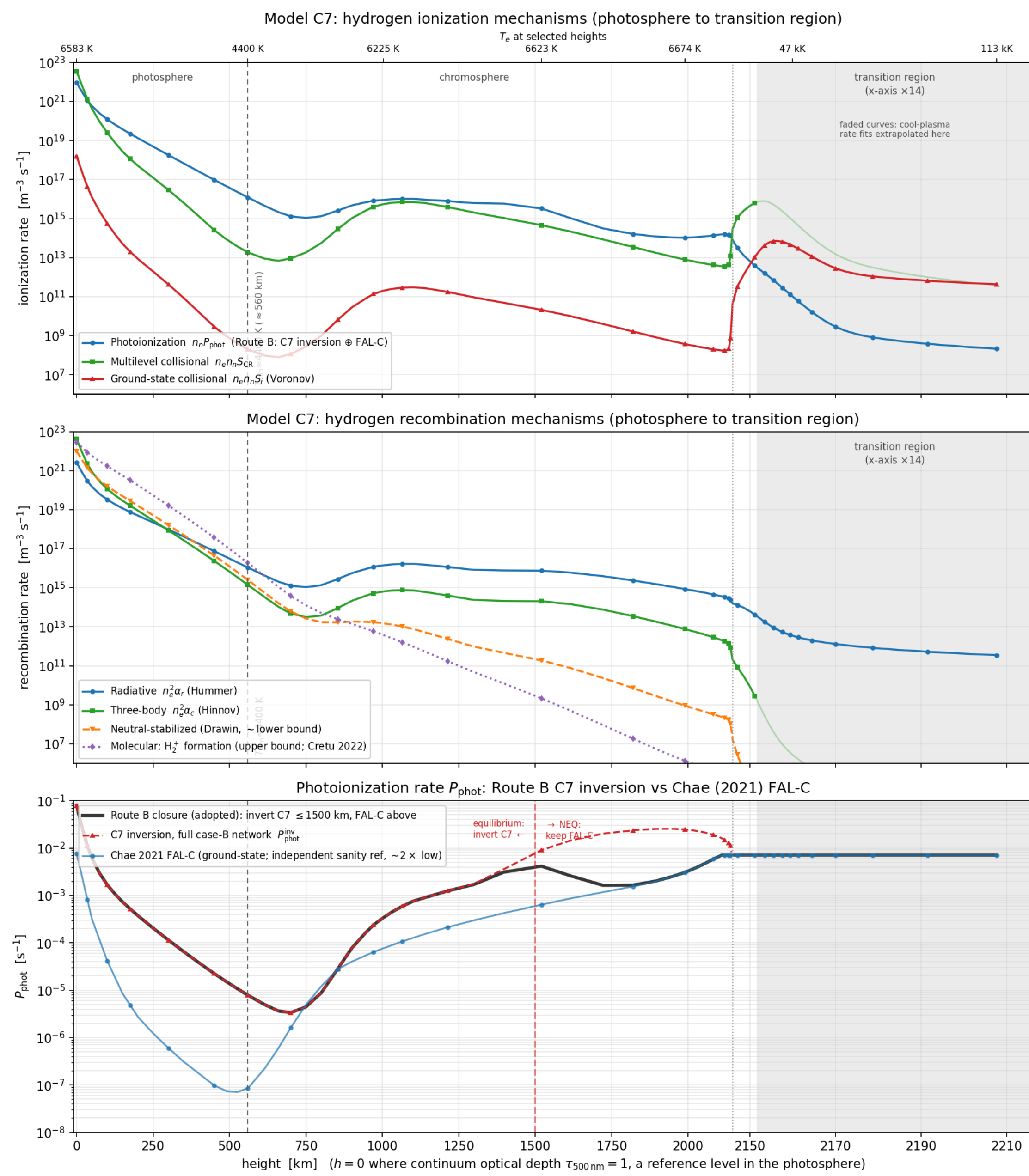
$$\frac{\partial \mathbf{U}_i}{\partial t} = -\frac{B_i}{\Delta s_i} \left( \frac{\mathbf{F}_{i+\frac{1}{2}}}{B_{i+\frac{1}{2}}} - \frac{\mathbf{F}_{i-\frac{1}{2}}}{B_{i-\frac{1}{2}}} \right) + \mathbf{S}_i$$

- **Space**: TVD–MUSCL limited-linear reconstruction → 2nd order.
- **Flux**: Rusanov (local Lax–Friedrichs).
- **Time**: 2-stage Runge–Kutta → 2nd order.
- **Stiff sources** operator-split, each advanced by the full  $\Delta t$ : transport  $\Rightarrow$  drag  $\Rightarrow T_i - T_n$  equilibration  $\Rightarrow$  conduction (Thomas)  $\Rightarrow$  ionization (cell quadratic)  $\Rightarrow$  cooling (backward–Euler, positivity-preserving).



**Finite-volume stencil**: cell-averaged states  $\mathbf{U}_i$  updated by interface fluxes  $\mathbf{F}_{i+1/2}$  along the field line.

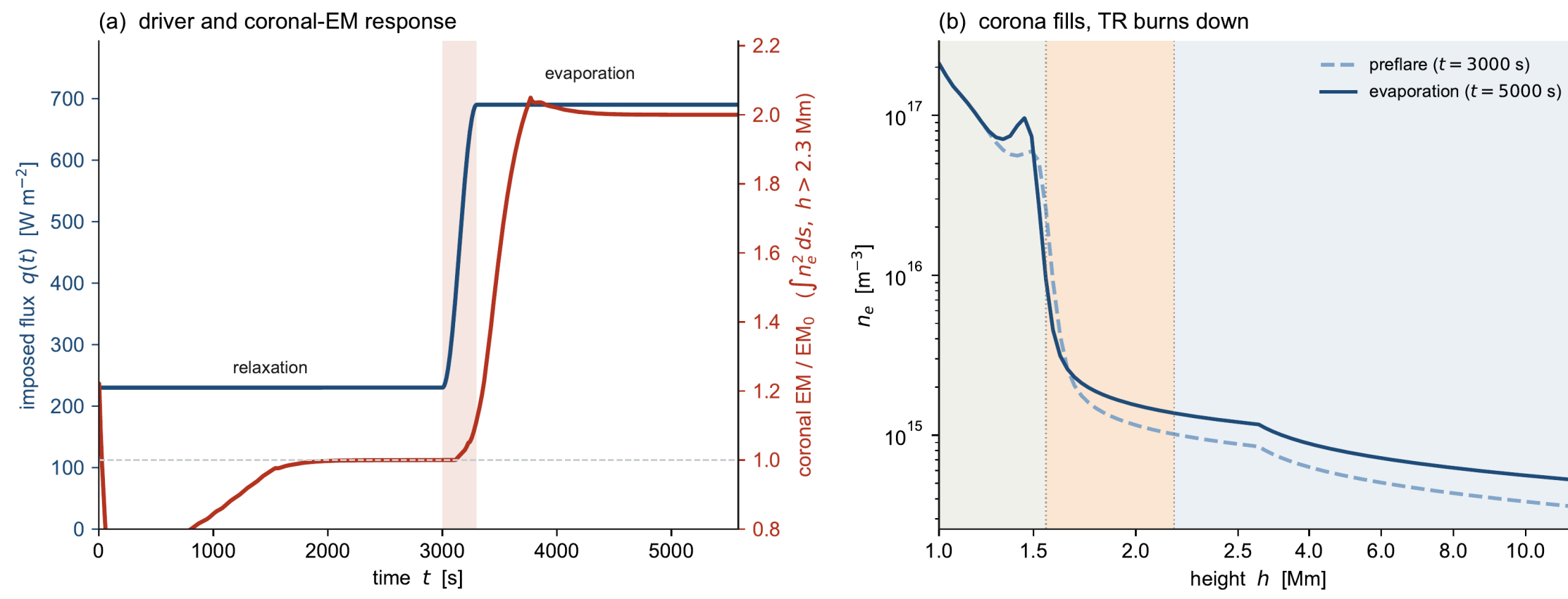
### Result 1 — Ionization & recombination in C7



**Dense base**: the collisional pair ( $S_{CR}$ ,  $\alpha_c \propto n_e$ ) dominates (Saha-like). **Mid/upper chromosphere**: photoionization + radiative recombination. **TR**: direct collisional  $S_i$ . Route-B closure matches FAL-C above 1500 km; column cooling  $\approx 6.6 \text{ kW m}^{-2}$  ( $\sim 10\%$  of the CL2012 Bifrost benchmark).

### Result 2 — Gentle (conduction-driven) evaporation

Resolved C7 corona to  $h \approx 10$  Mm; relax to a quiet steady state (apex  $\sim 1.1$  MK), then **ramp the coronal flux**  $q(T) \times 3$  over 300 s — **no electron beam**.



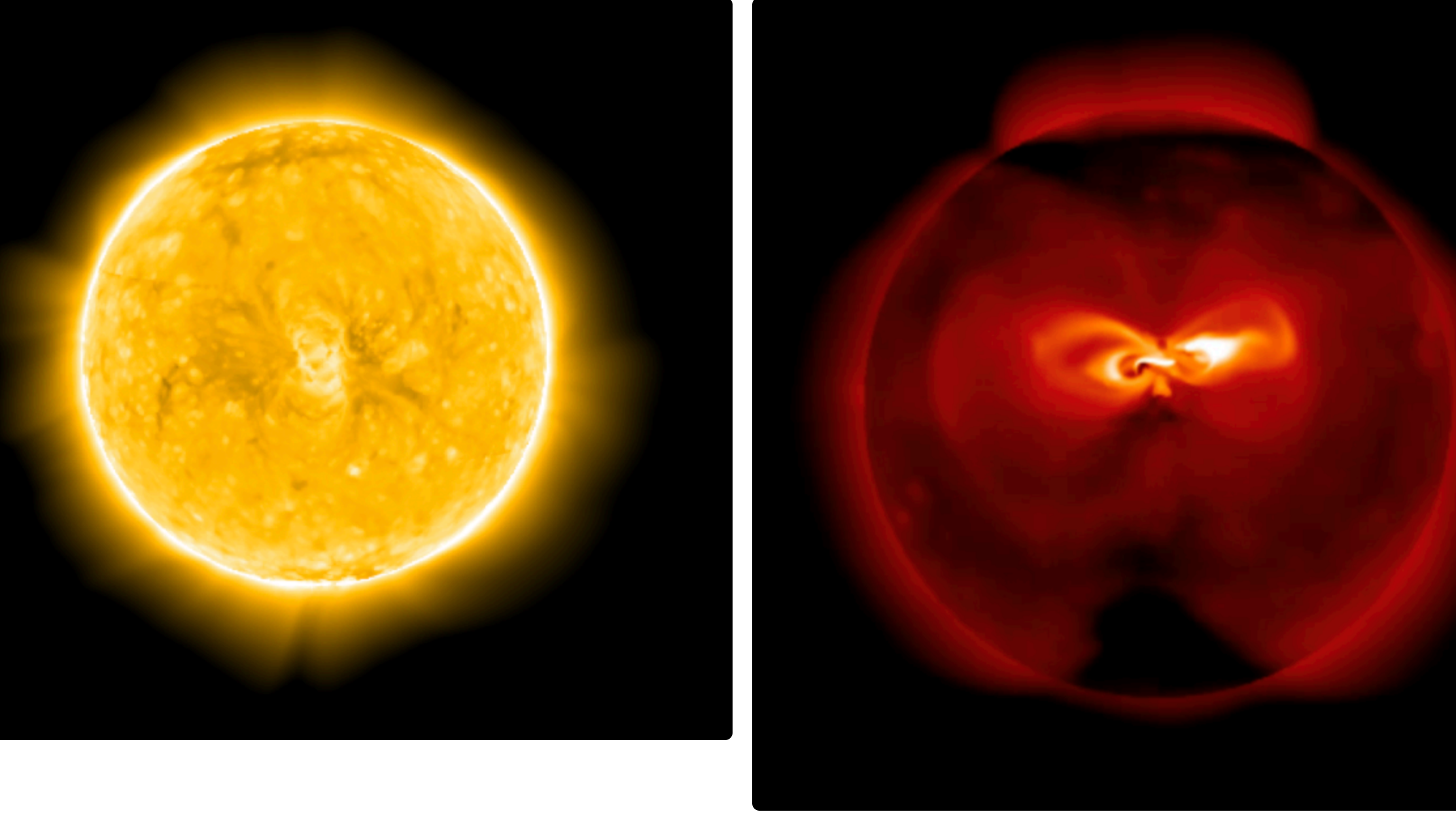
| Quantity                   | Quiet   | Evaporating        |
|----------------------------|---------|--------------------|
| Apex temperature           | 1.14 MK | 1.50 MK            |
| Coronal EM $\int n_e^2 ds$ | 1x      | 2x                 |
| Coronal column mass        | —       | +40%               |
| Peak upflow                | 0       | Mach $\approx 0.3$ |

Subsonic upflow  $\Rightarrow$  *gentle* evaporation (Antiochos & Sturrock 1978); the TR burns down and the resolved corona fills.

### Conclusions & outlook

**A self-consistent, dynamic chromosphere that solves the lower boundary AWSoM-R currently prescribes.**

- A field-aligned **two-fluid** model: non-equilibrium H ionization, multi-channel radiative cooling, optional separate  $T_e$ , physical coronal BC — run along **PFSS field lines**.
- Recovers chromospheric ionization structure *and* the gentle conduction-driven evaporation signature (EM rise, subsonic upflow, TR burndown).
- Designed as a **drop-in dynamic lower boundary for AWSoM-R**.
- **Next**: two-way coupling into SWMF; PFSS-line ensembles for full events.



Downstream science enabled: synthetic AIA 171 Å of a global coronal model (left) and a synthetic X-ray CME (right).

### Acknowledgements & key references

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